**Course Code: ITA1439** **Course Name: Ethical Hacking**

**Slot: A**

# CO1 – ASSIGNMENT_1

**Nature of Assignment: Individual** **Duration: 1 Week** **Total Marks: 50**

**Student Name:** **Reg. No:**

**CO1:** Apply the foundational concepts of ethical hacking, including the types of hackers, ethical principles, legal frameworks, and the role of responsible disclosure in cybersecurity.

**Assignment Title:** Design and Develop a Complete Ethical-Hacking Engagement Framework for a University's Authorized Security-Assessment Programme.

## Scenario - University Authorized Security-Assessment Programme

A university is establishing an Authorized Security-Assessment Programme to test the security of its student portal, learning-management system, email service and campus network before a major upgrade. The programme must cover approximately 20 systems and 50 staff initially, but the framework is expected to remain usable as the programme grows to the whole university and begins to include external, third-party assessors.

The management has not yet fixed how ethical-hacking engagements will be governed. Testing may be requested by different departments, carried out by internal staff or external testers, and may touch systems that hold personal data. The team is concerned that an unstructured approach may lead to unauthorized access, legal exposure, damage to live systems, unclear scope or uncoordinated disclosure of discovered vulnerabilities.

You have been appointed as the Ethical-Hacking Governance Team. Your responsibility is not merely to describe hacking tools. You must investigate the requirements, design a complete engagement framework, construct reusable components and documents, integrate the different parts, validate the design through a walkthrough and defend the final framework using evidence.

The final framework must clearly distinguish between what is authorized, what is legal, and what is appropriate for the university. The team is expected to make governance decisions independently and justify why the framework is suitable for the given scenario. No live exploitation is performed; all work stays at the design, documentation and isolated-lab level.

**Design Challenge**

Create and validate an integrated ethical-hacking engagement framework for the university programme. The framework must define how an engagement is authorized, how scope and rules of engagement are

set, how the relevant laws are respected, how different types of attackers are modelled, how discovered vulnerabilities are disclosed responsibly, and how an isolated lab is used where hands-on testing is required.

Your final design must be usable in practice and must demonstrate interaction among the selected components rather than treating each document or rule as an isolated example. Every major governance decision must be supported by evidence and a clear explanation of the trade-offs involved.

**TASK 1 – Formulate the Enquiry and Framework Blueprint** **4 Marks**

• Convert the scenario into a formal set of engagement-governance requirements. Identify the systems in scope, the stakeholders, the expected activities and the constraints.

• Create a framework blueprint showing how authorization, scope, legal compliance, threat modelling, lab testing and disclosure will cooperate in the proposed programme.

• Formulate at least four enquiry questions that your framework must answer, such as who authorizes a test, what is in scope, which laws apply and how findings are disclosed.

• Define at least five design criteria and use them as acceptance criteria for the final framework.

**Evidence Expected**

A requirements-to-governance blueprint, enquiry questions, design criteria and a clear rationale showing how the proposed framework addresses the scenario.

**TASK 2 – Design a Hacker Threat Model and Classification** **4 Marks**

• Design a classification of the types of hackers relevant to the university, such as white-hat, black-hat, grey-hat, script kiddie, hacktivist, insider and state-sponsored actors.

• Map each type to its motivation, typical capability and likely targets within the university.

• Establish clear criteria and naming so that an actor can be classified consistently.

• Demonstrate at least two alternative ways of classifying one borderline actor, then choose and justify the final classification.

**Evidence Expected**

A documented hacker threat model, classification criteria, a mapping table, rejected alternatives and a justified final classification.

**TASK 3 – Create an Ethics and Code-of-Conduct Module** **4 Marks**

• Define the ethical principles that govern an authorized engagement, such as written authorization, agreed scope, confidentiality, integrity, avoiding harm and honest reporting.

• Design rules or checks that help detect or prevent unethical actions during an engagement.

• Construct scenario cases containing acceptable, borderline and unacceptable actions and show how the code of conduct responds to each.

• Explain how the module preserves trust and keeps the engagement lawful and professional.

**Evidence Expected**

A reusable code-of-conduct module, decision rules, scenario test cases and evidence that the rules separate acceptable from unacceptable actions.

**TASK 4 – Construct a Legal-Compliance Mapping** **4 Marks**

• Map the engagement to the relevant provisions of the Information Technology Act, 2000 on unauthorized access and data, and compare them with an international law such as the Computer Fraud and Abuse Act.

• Build a table of offences, obligations and penalties related to unauthorized access and data protection.

• Connect each legal requirement to a specific control in the engagement framework.

• Decide and justify which legal requirements apply when a test touches systems that hold personal data.

**Evidence Expected**

A legal-compliance mapping, an offences-and-penalties table, the controls linked to each requirement and a justified conclusion on which laws apply.

**TASK 5 – Create a Scope and Rules-of-Engagement Document** **4 Marks**

• Design a scope and rules-of-engagement (RoE) document that defines in-scope and out-of-scope systems, testing windows, permitted and prohibited techniques and escalation paths.

• Create rules for recording and verifying the written authorization behind each engagement.

• Establish clear naming and boundaries so that an out-of-scope request can be detected.

• Construct at least one transformation of the RoE into a pre-engagement checklist.

**Evidence Expected**

A complete RoE document, authorization-verification rules, scope boundaries and a derived pre-engagement checklist.

**TASK 6 – Build an Authorization and Disclosure Layer** **5 Marks**

• Create an authorization workflow covering the request, approval, written sign-off and recording of each engagement.

• Design a responsible (coordinated) disclosure process with a reporting format, remediation window and timeline, and compare it with full disclosure and bug-bounty approaches.

• Develop operations for logging approvals and status updates and for deciding when disclosure may proceed.

• Create rules for choosing between coordinated and full disclosure for a given finding.

• Demonstrate that this layer interacts correctly with the rules-of-engagement and legal-compliance components.

**Evidence Expected**

An authorization workflow, a responsible-disclosure process, logging and decision operations, disclosure rules and evidence of interaction with the other components.

**TASK 7 – Design an Isolated Ethical-Hacking Lab** **5 Marks**

• Design a safe, isolated lab using a Kali Linux machine and one or more authorized target machines on a host-only network with reserved addresses such as 192.168.56.0/24.

• Specify the components, roles and connectivity of the lab and the steps used to record its configuration.

• Develop at least two operations, such as verifying connectivity between the machines and recording the lab setup.

• Construct a small comparison between a host-only and a NAT lab network and justify the chosen design.

• Explain how the lab keeps all testing authorized and contained within the isolated environment.

**Evidence Expected**

An isolated-lab design, a component and connectivity specification, verification operations, a comparison of network options and a justified final lab model.

**TASK 8 – Develop a Reusable Document Library** **5 Marks**

• Create a minimum of four reusable engagement documents supporting the framework, including at least one authorization form and one finding report.

• Use sensible default fields where appropriate and explicitly design the content of each document.

• At least one document, such as the finding report, must produce a structured output that integrates with another component of the framework.

• Create test entries that demonstrate valid, boundary and invalid inputs and document the expected result.

**Evidence Expected**

A reusable document library, documented fields, a structured report output, test entries and evidence of integration with the framework.

**TASK 9 – Integrate, Test and Refine the Framework** **5 Marks**

• Integrate the components created in Tasks 2-8 into one coherent engagement framework.

• Construct a realistic walkthrough of an authorized assessment from request and authorization, through testing in the isolated lab, to responsible disclosure.

• Design systematic test cases covering missing authorization, out-of-scope requests, legal conflicts and delayed disclosure.

• Record the problems encountered during the walkthrough and show how the framework was refined to address them.

**Evidence Expected**

A reproducible integrated framework, a workflow walkthrough, systematic test cases, refinement evidence and a final corrected framework.

**TASK 10 – Create a Scalability and Governance Evaluation 5 Marks**

• Construct scenarios representing a small programme (one department) and a large programme (the whole university with external assessors), with the number of systems and testers involved.

• Experiment with at least three important processes from the framework and record meaningful measures such as approval turnaround, scope coverage or the number of steps.

• Create a comparison table and at least one suitable chart or structured visualization.

• Use the evidence to refine the governance of the framework rather than simply reporting the measures.

**Evidence Expected**

A reproducible evaluation setup, a comparison table, a visualization, an interpretation and evidence of at least one governance refinement based on the results.

**TASK 11 – Defend the Framework Under a Changed Requirement 5 Marks**

• Changed requirement: the university now runs continuous and third-party assessments with stricter data-protection rules and a shorter disclosure timeline, and needs the framework to remain easy to govern and extend.

• Create a revised version of your framework by deciding which components should be retained, modified, replaced or redesigned.

• Construct at least one new or modified component that directly addresses the changed requirement and show how it integrates with the existing framework.

• Defend the final framework using evidence from your walkthrough, scalability evaluation, legal mapping and disclosure process.

• Identify at least two trade-offs or limitations that remain in your final design and propose future improvements.

**Evidence Expected**

A revised framework, demonstrated adaptation, an evidence-based defence, explicit trade-offs and a clear explanation of how the final design satisfies the changed requirements.

**MARK DISTRIBUTION**

| Task | Marks |
|---|---|
| TASK 1 – Formulate | 4 |
| TASK 2 – Design | 4 |
| TASK 3 – Create | 4 |
| TASK 4 – Construct | 4 |
| TASK 5 – Create | 4 |
| TASK 6 – Build | 5 |
| TASK 7 – Design | 5 |
| TASK 8 – Develop | 5 |
| TASK 9 – Integrate | 5 |
| TASK 10 – Create | 5 |
| TASK 11 – Defend | 5 |
| **TOTAL** | **50** |